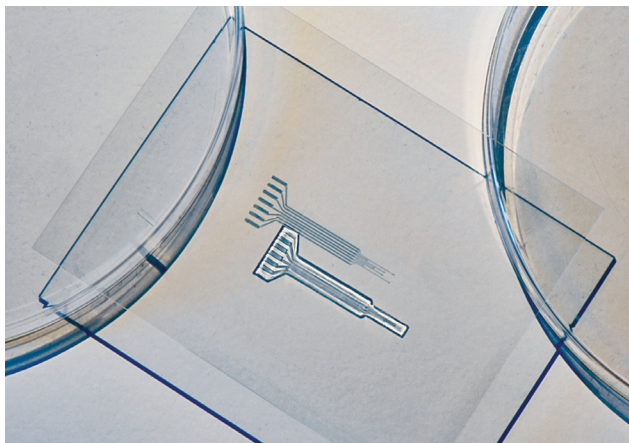


Soft, printable, metal-free electrodes for electronic implants

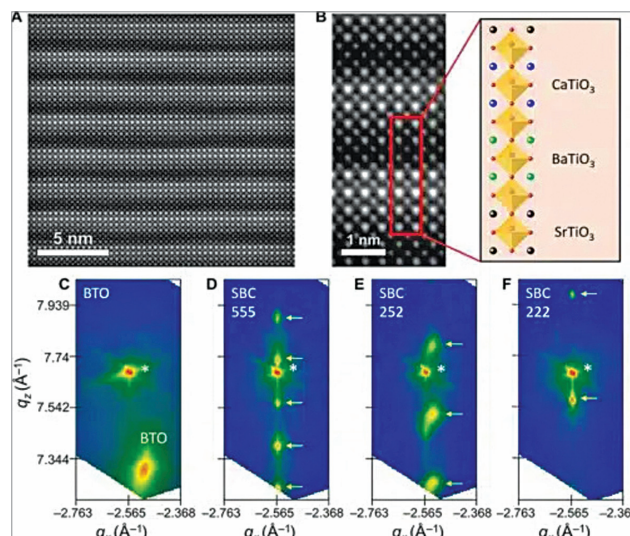


MIT engineers developed a metal-free, Jell-O-like material that is as soft and tough as biological tissue and can conduct electricity similarly to conventional metals. The new material, which is a type of high-performance conducting polymer hydrogel, may one day replace metals in the electrodes of medical devices (Credit: Felice Frankel)

MIT engineers have created a metal-free, gelatinous substance that shares biological tissue's softness and resilience while exhibiting electrical conductivity similar to traditional metals. The researchers used conductive polymers to make metal-free electrodes for bioelectronic implants, aiming for a flexible, biocompatible, conductive gel by combining conductive polymer and hydrogel. The team developed a conductive polymer and hydrogel blend, improving their electrical and mechanical properties. Mixing them created dispersed polymer particle gels. Using a 3D printer, they printed electrode patterns on hydrogel films. Implanted on rats' hearts, nerves, and spinal cords, the electrodes performed well for two months with minimal inflammation. They transmitted electrical pulses, stimulating motor activity effectively. The team aims to enhance the gel's durability and functionality for long-term implants, such as pacemakers and deep-brain stimulators, enabling seamless organ connections.

Hyper-efficient solar panels: 1000x more powerful

Researchers at Martin Luther University Halle-Wittenberg (MLU) have discovered a new method to boost solar cell efficiency by 1,000 times. The researchers vapourised crystals with a laser, embedding barium titanate between strontium titanate and calcium titanate on substrates. The resulting material has 500 layers, around 200 nanometres thick. Laser irradiation during photoelectric measurements increased current flow up to 1,000 times compared to pure barium titanate. Lattice layers boost permittivity, facilitating electron flow through light excitation. The layered structure improves temperature performance, durability, and eliminates the need for special packaging. The new material boosts solar panel efficiency, making it ideal for limited urban spaces. It enhances the photovoltaic effect of ferroelectric crystals, improving solar cell efficiency. The team aims to build a prototype solar cell, with the potential for future commercial panels using the new material.



Structural characterisation of superlattices. (A) Cross-sectional STEM acquired from sample SBC222. (B) High-resolution STEM from a part of the scanned region. The schematic depicts the arrangement of unit cells. RSM acquired around (103) reflection in (C) BTO, (D) SBC555, (E) SBC252, and (F) SBC222. Star and yellow arrows indicate the STO substrate and satellite peaks from SL, respectively (Credit: Uni Halle/Yeseul Yun)